

Task 3: Data Visualization

Objective

The objective of this task was to transform raw data into meaningful visual representations using Python libraries such as Matplotlib and Seaborn. Data visualization helps in understanding patterns, trends, relationships, and insights within the dataset.

Dataset Used

For this project, the "Tips" dataset available in the Seaborn library was used. The dataset contains information about restaurant bills, tips, customer gender, smoking habits, day, time, and group size.

Tools and Libraries

- Python
- Google Colab
- Pandas
- Matplotlib
- Seaborn

Visualizations Created

1. Bar Chart

A bar chart was created to visualize the distribution of customers based on gender. The chart showed that male customers were more frequent than female customers.

2. Pie Chart

A pie chart was used to represent the percentage distribution of customer genders. This visualization provided a clear understanding of the proportion of male and female customers.

3. Histogram

A histogram was created to analyze the distribution of total bill amounts. Most bill amounts were concentrated between \$10 and \$30.

4. Scatter Plot

A scatter plot was used to study the relationship between total bill amount and tip amount. The plot indicated a positive relationship, suggesting that customers generally gave higher tips for higher bills.

5. Box Plot

A box plot was used to identify the presence of outliers in the total bill data. The visualization showed a few higher bill values but no significant anomalies.

6. Heatmap

A correlation heatmap was generated to understand relationships between numerical variables. The heatmap revealed a positive correlation between total bill and tip amount.

7. Dashboard

A simple dashboard combining multiple charts was created to provide a comprehensive overview of the dataset in a single view.

Key Insights

1. Male customers visited the restaurant more frequently than female customers.
2. Most bills ranged between \$10 and \$30.
3. Higher bill amounts generally resulted in higher tips.
4. The dataset showed a positive correlation between total bill and tip amount.
5. No major data quality issues or extreme outliers were observed.

Conclusion

Data visualization successfully transformed raw data into meaningful insights. Various charts and graphs helped identify patterns, trends, and relationships within the dataset. This task improved understanding of data storytelling and visual analytics, which are essential skills in data analytics and decision-making.